

Jiahui (Clay) Yang

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R&D 100 Award Winner (2025)

[Personal Website](#)

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TECHNICAL PROFILE

Ph.D. student in Electrical Engineering specializing in **PMU hardware–algorithm co-design, frequency & synchrophasor estimation**, oscillation detection, and [wide-area monitoring deployments across the U.S. and internationally](#). Experience also includes FPGA-based high-precision timing source development for PPS synchronization. Proven ability to solve practical challenges in power systems via **Python, C++, LLM APIs**.

EDUCATION

Ph.D. in Electrical Engineering University of Tennessee, Knoxville	Aug. 2024 - Present
<i>Coursework (in progress): Advanced Large Language Models (Reasoning, Agents, and the Structure of Intelligence)</i>	
M.S. in Electrical Engineering Tianjin University	Sep. 2021 – Jul. 2024
B.S. in Electrical Engineering and Automation Tianjin University	Sep. 2017 – Jul. 2021

PROJECTS

- [FNET/GridEye](#) Power System Monitoring (UTK & ORNL)**
 - ✧ **Led the development, deployment, and maintenance of hundreds of PMUs** across the U.S. and internationally under the supervision of Dr. Yilu Liu, using FnetServer & OpenPDC.
 - ✧ Proficient in **PMU and waveform hardware**, including PCB design & manufacturing, ADC sampling, GPS/PPS timing, and IEEE C37.118-compliant data transmission.
 - ✧ Upgraded and implemented **real-time frequency and synchrophasor estimation algorithms**.
 - ✧ **Leveraged LLM APIs** to build a device-management web portal (unit metadata, ticketing, status dashboard, data-quality checks) and an automated email assistant for support workflows.
 - ✧ Evaluated P-class and M-class PMUs for **0.1-30 Hz oscillation detection**
- Pulsar Signal-based High-Precision Timing Source**
 - ✧ **Developed FPGA pipeline in Simulink** to process high-rate ADC samples in real time and stream data by 10-GbE interface and PF_RING package using Python and C++.
 - ✧ Developed polyphase filter bank, folding, and period estimation algorithms to extract pulsar period, generate PPS synchronized to GPS and distribute timing via **White Rabbit**.
- Key Technologies of Digital Power Grid**
 - ✧ Collaborated with **China Southern Power Grid** to develop edge-computing devices for distribution grids, including containerized data compression and local voltage-control apps.

PUBLICATIONS

- First Author, IEEE Transactions on Power Systems, 2024.** Real-time D-PMU data compression for edge computing devices in digital distribution networks.
- Corresponding Author, IEEE Transactions on Industry Applications, 2025.** Lightweight data-driven planning method of hybrid energy storage systems in the new power system.
- IEEE EFTF/IFCS, 2025. A hybrid system for high-accuracy timekeeping using millisecond pulsars.

SKILLS

Programming: LLM API integration, C/C++, Python, MATLAB.

Modeling & Simulation: Simulink, PSS®E, PSCAD.

Embedded system: Raspberry Pi, STM32, DSP, FPGA, ESP32, u-blox (GPS), Arduino.

Software: Altium/KiCad, Keil, Git; AWS, Azure, Google Cloud; Microsoft Office